Lab 5 – Comparing Health Risks

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Formatting Instructions



- Please submit your lab report as a **pdf** to Gradescope.
- When you upload to Gradescope, please **match pages** with the **question number**.
- Be sure that all **group members** are **added** in your submission to Gradescope (click view/edit group on the top right of the page once shown your final submission after matching pages).

Assignment Overview

- We'll be investigating the heart dataset, which collected data on the health factors of 303 patients being screened for heart disease. We'll use this data to address the following three research questions (one on each page):
    - Do people with fasting blood sugar levels above 120 mg/dL have a **higher** risk for heart disease?
    - Do people who have experienced an exercise induced angina have a **higher** risk for heart disease?
    - Do people who experience exercise induced anginas have **different** cholesterol levels on average?

STEP 0

- **Pre-lab work**
    - Complete the pre-lab tutorial (Comparing Groups) for Lab 5 first: https://stat212-learnr.stat.illinois.edu/
- **Download** the heart.csv file to your computer and then **import** into your RStudio session.
- Create a new **R script** (or use the **RMarkdown file** if you are using that option)
- Remember to **library(tidyverse)** so that you can use the ggplot function!

Variables

Each row of this dataset represents one patient being screened, and the following variables were documented for each patient:

- **age:** age in years
- **sex:** biological sex (0 if female, 1 if male)
- **cp**: chest pain type (0 if typical angina, 1 if atypical angina, 2 if non-anginal pain, 3 if asymptomatic)
- **exang:** binary variable documenting whether patient experienced exercise induced angina
- **trestbps:** resting systolic blood pressure (in mm/Hg on admission to hospital)
- **chol:** serum cholesterol (mg/dL)
- **fbs:** binary variable documenting whether fasting blood sugar was high ("yes" if > 120 mg/dL and "no" if <= 120 mg/dL)
- **restecg:** resting electrocardiographic results (0 if normal, 1 if having ST-T wave abnormality, 2 if showing probable or definite left ventricular hypertrophy)
- **thalach:** maximum heart rate achieved
- **oldpeak:** ST depression induced by exercise relative to rest
- **slope:** the slope of the peak exercise ST segment
- **ca:** number of major vessels (0-3) colored by flourosopy
- **target:** Whether patient was found to have angiographic disease status (heart disease) as determined by amount of blood vessel narrowing ("positive" if heart disease diagnosis, "negative" if no heart disease diagnosis)

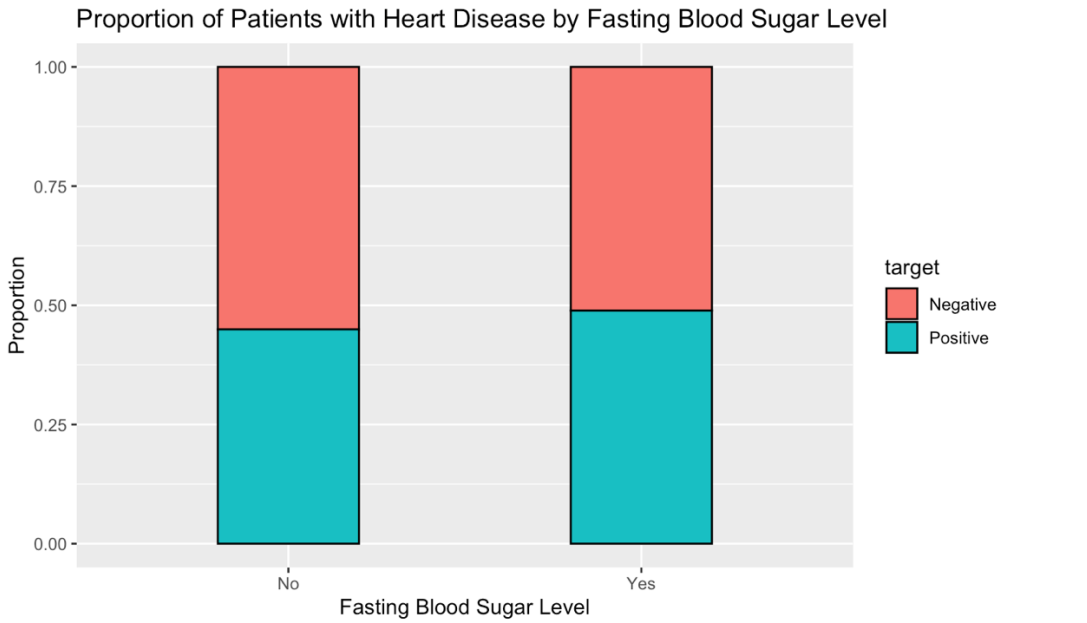
Research Question 1: *Do people who are diabetic (fasting blood sugar levels above 120 mg/dL) have a* ***higher*** *risk for heart disease?*

Question 1 (5pts): Let's first investigate visually. **Create a 100% stacked barplot** to compare the proportion of patients with heart disease based on whether their fasting blood sugar level was above 120 mg/dL.

Include an image of your barplot in the report *and* **Include your R code**

- One bar should represent those who are diabetic, and the other should represent those who are not. The bar should be shaded to reflect what proportion in each group have heart disease.
- Give the bars a black border, and adjust the width to be between 0.2 and 0.5
- Add an appropriate x axis label, y axis label, and title.

```
75
76 ```{r}
77 ggplot(heart, aes(x = fbs, fill = target)) +
78   geom_bar(position = "fill", width = 0.4, color = "black") +
79   labs(x = "Fasting Blood Sugar Level", y = "Proportion", title = "Proportion of Patients with Heart
   Disease by Fasting Blood Sugar Level")
80 ```
```



Question 2 (5pts): Now, let's use a test for two proportions to make a statistical inference. Using the dplyr package, create a frequency table to get counts of how many people have or don't have heart disease based on whether they are diabetic or not.

Copy or screenshot the frequency table into your report *and* **Include your R code**

- If done correctly, this table will have 4 rows.
- You can display the table exactly as it appears in R output, or you can re-format it in your document if you wish to.

Run a proportions test to answer research question 1 *and* **Include your R code.**

- **Tip:** Is this a directional or non-directional test? Read the research question again!
- Remember that you need to enter two vectors into your code, the first vector includes the numbers in each group who have heart disease, and the second vector includes the totals for each group.
- Copy+paste or screenshot the summary output from your proportions test.

In your own words, interpret the results and make a conclusion in context. A full response should:

- Identify the proportion with heart disease in each group
- Identify the p-value
- Briefly summarize your answer to our first research question using these results.

```
103
104 ```{r}
105 heart %>%
106   group_by(fbs, target) %>%
107   summarise(count = n(), .groups = "drop")
108
109 #prop.test(x = c(116, 22), n = c(116 + 142, 22 + 23), alternative = "greater")
110 ```
```

A tibble: 4 × 3

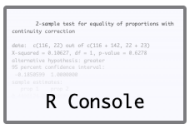
fbs <chr>	target <chr>	count <int>
No	Negative	142
No	Positive	116
Yes	Negative	23
Yes	Positive	22

4 rows

```r
heart %>%
  group_by(fbs, target) %>%
  summarise(count = n(), .groups = "drop")

prop.test(x = c(116, 22), n = c(116 + 142, 22 + 23), alternative = "greater")
```






```
        2-sample test for equality of proportions with continuity correction

data:  c(116, 22) out of c(116 + 142, 22 + 23)
X-squared = 0.10627, df = 1, p-value = 0.6278
alternative hypothesis: greater
95 percent confidence interval:
 -0.1850599  1.0000000
sample estimates:
   prop 1    prop 2
0.4496124 0.4888889
```

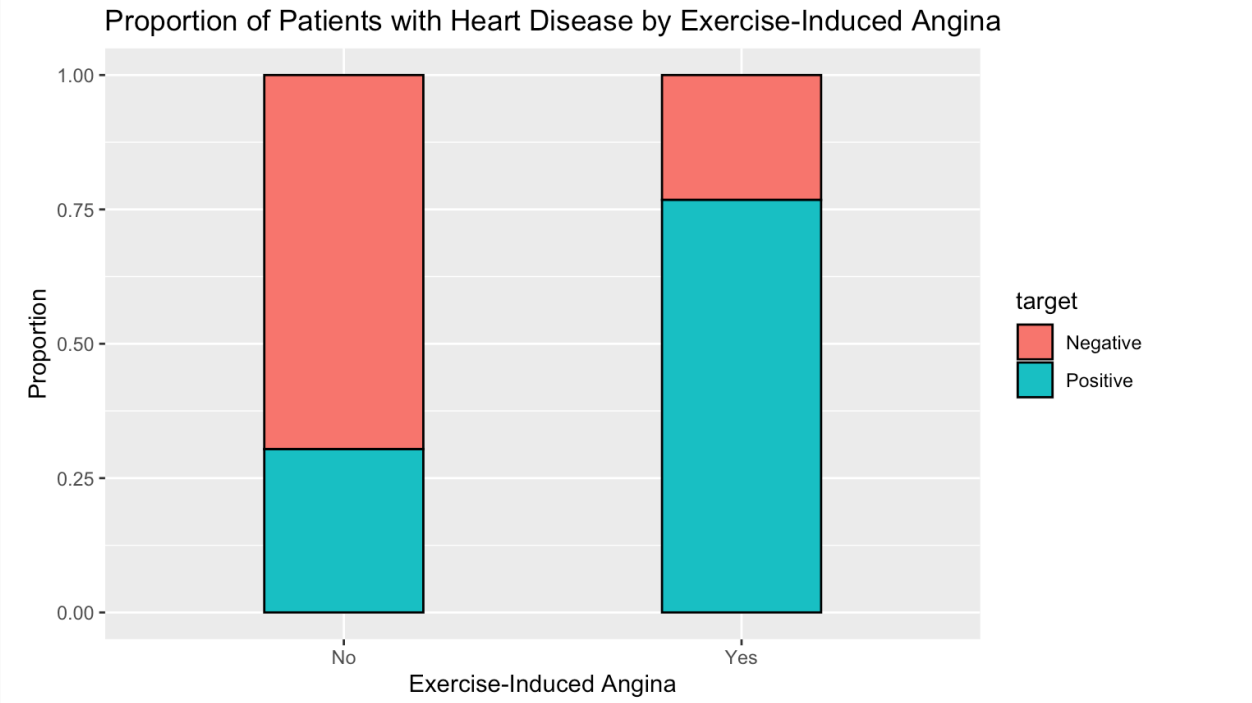
This is a directional test. The results shows that there is not enough evidence to conclude that people with heart disease are greater in the diabetic group compared to the non-diabetic group. The p-value of 0.6278 is relatively high and not statistically significant at a typical significance level of 0.05.

Research Question 2: *Do people who have experienced an exercise induced angina have a* ***higher*** *risk for heart disease?*

Question 3 (5pts): Repeat the procedures for Question 1, but with this new predictor variable.

Include an image of your 100% stacked barplot in the report *and* **Include your R code**

```
122
123 ```{r}
124 ggplot(heart, aes(x = exang, fill = target)) +
125   geom_bar(position = "fill", color = "black", width = 0.4) +
126   labs(x = "Exercise-Induced Angina", y = "Proportion", title = "Proportion of Patients with Heart
    Disease by Exercise-Induced Angina")
127 ```
```



Question 4 (5pts): Follow the same procedures in Question 2 to address our second research question statistically.

Copy or screenshot the frequency table into your report *and* **Include your R code**

Run a proportions test designed to answer your second research question *and* **Include your R code.**

In your own words, interpret the results and make a conclusion in context (same as Question 2).

```r
heart %>%
  group_by(exang, target) %>%
  summarise(count = n(), .groups = "drop")

prop.test(x = c(62, 76), n = c(62 + 142, 76 + 23), alternative = "greater")
```

tbl_df 4 x 3 | R Console

A tibble: 4 × 3

exang <chr>	target <chr>	count <int>
No	Negative	142
No	Positive	62
Yes	Negative	23
Yes	Positive	76

4 rows

```r
heart %>%
  group_by(exang, target) %>%
  summarise(count = n(), .groups = "drop")

prop.test(x = c(62, 76), n = c(62 + 142, 76 + 23), alternative = "greater")
```

tbl_df 4 x 3 | R Console

```
        2-sample test for equality of proportions with continuity correction

data:  c(62, 76) out of c(62 + 142, 76 + 23)
X-squared = 55.945, df = 1, p-value = 1
alternative hypothesis: greater
95 percent confidence interval:
 -0.558891  1.000000
sample estimates:
   prop 1    prop 2
0.3039216 0.7676768
```

These results suggest a significant difference in people with heart disease between those who have experienced exercise-induced angina and those who haven't. The p-value is extremely low (1.0), which indicates strong evidence to support that people with heart disease is greater in the group that has experienced exercise-induced angina.

Question 5 (5pts): Let's now report the odds ratio for heart disease for each set of two groups we're comparing. Rather than code this computationally in R, we will use an **online calculator**! *We'll talk more about this in Chapter 9, but and Odds Ratio is similar to Relative Risk, but is more appropriate for this design context.*

Report the odds ratio (and 95% confidence interval) for heart disease when patient is diabetic (fasting blood sugar is above 120 mg/dL) as compared to when they are not diabetic.

Report the odds ratio (and 95% confidence interval) for heart disease when the patient had experienced an exercise induced angina as compared to one who didn't.

"Simpler" calculator suggestion: https://www2.ccrb.cuhk.edu.hk/stat/confidence%20interval/CI%20for%20ratio.htm

- *"Feature Present" row should represent number of patients with causal factor (fbs above 120, or angina reported), and outcome positive column should represent number who experience adverse outcome (heart disease)*
- *Be sure to set the 1 – α = 0.95 for a 95% confidence interval*

Calculator that Kelly used in class: https://istats.shinyapps.io/Association_Categorical/

- *Choose contingency table setup*
- *Have your rows represent your explanatory variables (fbs above/below 120, or angina status), and columns represent number who experience adverse outcome (heart disease)*
- *Be sure to choose odds ratio from the drop-down and choose the "95% confidence interval" option below.*

Test Group	Outcome Positive	Outcome Negative	Totals
Feature Present	22	23	45
Feature Absent	116	142	258
Totals	138	165	303

1–α: 0.95 | 重置 | Calculate

Result:

Group	Proportion of Outcome Positive	Proportion of Outcome Negative
Feature Present	0.4888	0.51111
Feature Absent	0.44961	0.5503

Result	Point Estimate	Lower C.I.	Upper C.I.
Odds Ratio	1.17091	0.6212	2.20707
Absolute Risk Reduction*	-0.03928	-0.19744	0.11889
Relative Risk Reduction**	-0.08736	-0.5092	0.21658
Number Needed to Treat***	-25.4605	-5.06483	8.41136
Patient Expected Event Rate	0.44961		

Test Group	Outcome Positive	Outcome Negative	Totals
Feature Present	76	23	99
Feature Absent	62	142	204
Totals	138	165	303

1–α: 0.95 | 重置 | Calculate

Result:

Group	Proportion of Outcome Positive	Proportion of Outcome Negative
Feature Present	0.7676	0.2323
Feature Absent	0.3039	0.6960

Result	Point Estimate	Lower C.I.	Upper C.I.
Odds Ratio	7.56802	4.35026	13.1658
Absolute Risk Reduction*	-0.46376	-0.56818	-0.35933
Relative Risk Reduction**	-1.5259	-2.19263	-0.99841
Number Needed to Treat***	-2.15631	-1.76001	-2.78294
Patient Expected Event Rate	0.30392		

Enter Data:

Contingency Table

Row Variable: fbs above/below 120

Category Labels: Diabetic (fbs>120), Non-D

Column Variable: heart disease

Category Labels: Positive, Negative

Enter Counts for Contingency Table:

	Positive	Negative
Diabetic (fbs>12(	22	23
Non-Diabetic (fbs	116	142

Statistic for Describing Association:

Odds Ratio

Proportions refer to:

1st Column 2nd Column

Show 95% Confidence Interval

Select Type of Chart:

Side-by-Side Bar Graph

Modify Labels of:

X-axis Y-axis

Title Subtitle

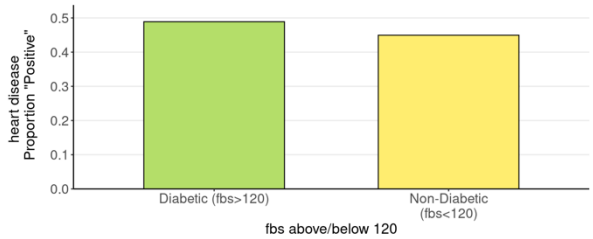
Contingency Table (Observed Counts):

fbs above/below 120	heart disease Positive	Negative	Total
Diabetic (fbs>120)	22	23	45
Non-Diabetic (fbs<120)	116	142	258
Total	138	165	303

Conditional Sample Proportions:

fbs above/below 120	heart disease Positive	Negative	Total
Diabetic (fbs>120)	0.4889	0.5111	1
Non-Diabetic (fbs<120)	0.4496	0.5504	1
Total	0.4554	0.5446	1

Bar Graph of Conditional Proportions



Descriptive Statistics:

Odds of "Positive" for "Diabetic (fbs>120)"	Odds of "Positive" for "Non-Diabetic (fbs<120)"	Odds Ratio
0.957	0.817	1.171

Inferential Statistics:

Population Parameter	Estimate	95% Confidence Interval
Odds Ratio	1.171	(0.624, 2.196)

Enter Data:

Contingency Table

Row Variable: angina status

Category Labels: angina, Non-angina

Column Variable: heart disease

Category Labels: Positive, Negative

Enter Counts for Contingency Table:

	Positive	Negative
angina	76	23
Non-angina	62	142

Statistic for Describing Association:

Odds Ratio

Proportions refer to:

1st Column 2nd Column

Show 95% Confidence Interval

Select Type of Chart:

Side-by-Side Bar Graph

Modify Labels of:

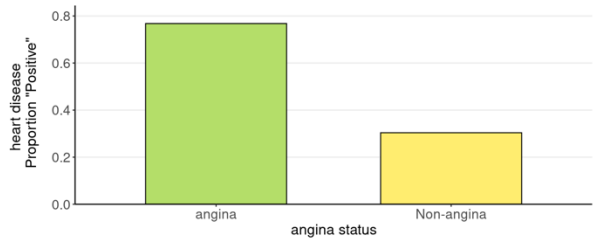
Contingency Table (Observed Counts):

angina status	heart disease Positive	Negative	Total
angina	76	23	99
Non-angina	62	142	204
Total	138	165	303

Conditional Sample Proportions:

angina status	heart disease Positive	Negative	Total
angina	0.7677	0.2323	1
Non-angina	0.3039	0.6961	1
Total	0.4554	0.5446	1

Bar Graph of Conditional Proportions



Descriptive Statistics:

Odds of "Positive" for "angina"	Odds of "Positive" for "Non-angina"	Odds Ratio
3.3	0.437	7.568

Inferential Statistics:

Population Parameter	Estimate	95% Confidence Interval
Odds Ratio	7.568	(4.36, 13.14)

Odds ratio (Diabetic vs. Non-Diabetic) = 1.17091

Odds ratio (Exercise-Induced Angina vs. No Exercise-Induced Angina) = 7.56802

Let's consider possible risk factors for high levels of cholesterols. Notice that cholesterol will be a *numeric* variable, so our approach to this question will be slightly different.

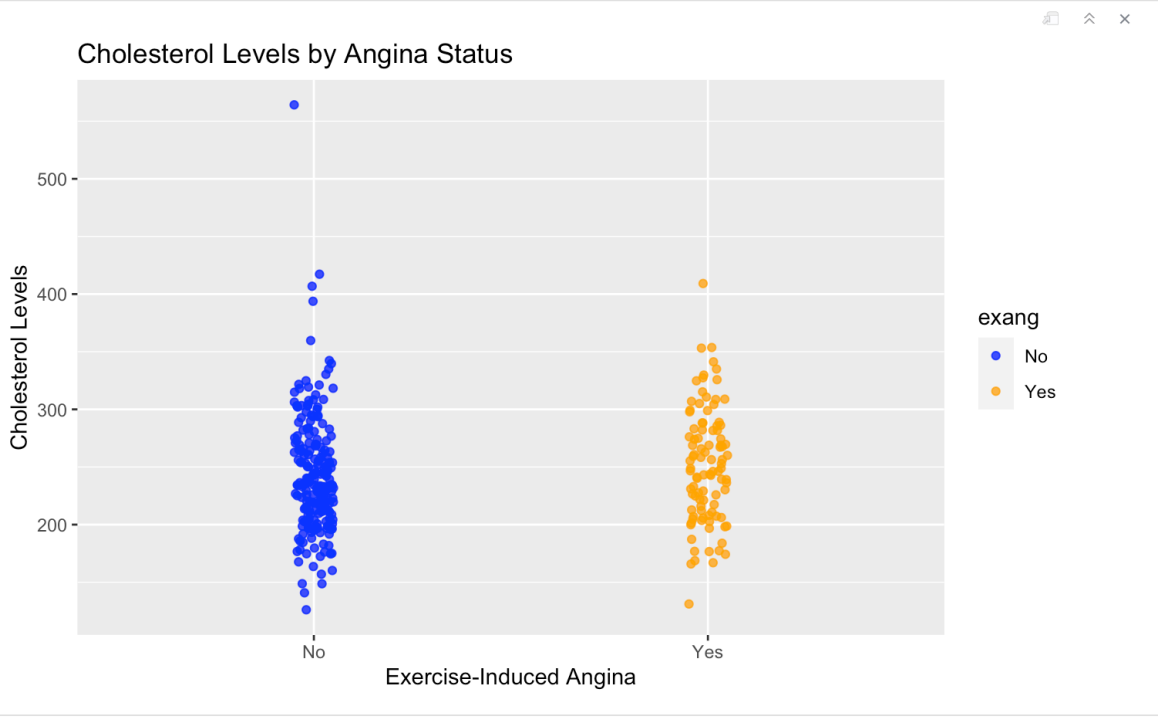
Research Question 3: *Do people who experience exercise-induced anginas have* ***different*** *cholesterol levels on average? Let's say the researchers believe either a drop or an increase in cholesterol is possible and noteworthy to report!*

Question 6 (5pts): **Create a jittered plot** to compare cholesterol levels between the angina and no angina groups.

Include an image of your jittered plot in the report *and* **Include your R code**

- Keep the width of your jitter small (like between 0.02 and 0.10)
- Color each group of points differently (one color for "No" and one color for "Yes")
- Add an appropriate x axis label, y axis label, and title

```r
```{r}
ggplot(heart, aes(x = exang, y = chol, color = exang)) +
  geom_jitter(width = 0.05, alpha = 0.8) +
  labs(x = "Exercise-Induced Angina", y = "Cholesterol Levels", title = "Cholesterol Levels by Angina Status") +
  scale_color_manual(values = c("No" = "blue", "Yes" = "orange"))
```
```



Question 7 (5pts): **Complete a t-test** to address the research question posed. *Even though we have enough observations such that a z-test would be fine, it's easier in R to just run a t-test, and the results will be approximately the same! We will* ***not*** *assume equal variances (software can handle this situation easier, and this is the "safer" testing option).*

Copy or screenshot the summary output from your t-test

In your own words, interpret the results and make a conclusion in context. A full response should:

- Identify the average cholesterol level for each group
- Identify the p-value
- Briefly summarize how this result helps you address research question 3.

```
210
211 ```{r}
212 t.test(data = heart, chol ~ exang, var.equal = FALSE)
213 ```
```

```
        Welch Two Sample t-test

data:  chol by exang
t = -1.1929, df = 206.28, p-value = 0.2343
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 -19.615616   4.826846
sample estimates:
 mean in group No mean in group Yes
         243.8480          251.2424
```

The average cholesterol level in the "No Exercise-Induced Angina" group is approximately 243.85.

The average cholesterol level in the "Exercise-Induced Angina" group is approximately 251.24.

P-Value: The p-value associated with the t-test is approximately 0.2343.

The results indicate that there is no statistically significant difference in average cholesterol levels between people who experience exercise-induced angina and those who do not. The p-value of 0.2343 is greater than a typical significance level (0.05), suggesting that we do not have enough evidence to conclude that there is a significant difference in cholesterol levels between these two groups.